Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

PPH805 – ELECTROMAGNETIC THEORY

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. Write down the Maxwell's equations for a conducting medium.
2. Define poynting vector.
3. State snell's law of refraction.
4. Define polarizing angle.
5. Give the cutoff frequency of TE and TM waves.
6. What is a rectangular waveguide?
7. What is radiation resistance?
8. What is a Hertzian dipole antenna?
9. Define four vector.
10. Give the Maxwell's equation in covariant tensor form.

SECTION - B (5X5=25)

Answer **ALL** Questions

11. a) Explain how the concept of displacement current was introduced by Maxwell's to account for the production of magnetic field is empty space.

(or)

b) State and derive continuity equation.

12. a) Write down the boundary conditions at the surface of

discontinuity.

(or)

b) Find the electric field vector E perpendicular to the plane of incidence.

13. a) Give the charactenstics of TE waves.

(or)

b) Show that in a wave guide TEM waves can not be propagated along the axis.

14. a) Find the effective length of a half wave dipole.

(or)

b) Give the characteristics of Radio waves.

15. a) Show that D' Alembert's operator is invariant under Lorentz transformation.

(or)

b) Show that the space time interval is invariant under Lorentz transformation.

SECTION – C (3X10=30)

Answer any **THREE** Questions

16. State and prove poynting's theory and discuss the physical meaning of each term.

17. Discuss the reflection and refraction of electromagnetic waves at the interface of non-conducting media.

18. Explain the propagation of waves in a rectangular waves guide and deduce the relation.

19. Explain radian from Hertzian dipole antenna and deduce the relation for electric and magnetic field.

20. Derive the transformation equations for the electromagnetic potentials A and ϕ.

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